

Learning target DF3, version 2

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $h(y) = -4 \ln(y) + 3 \sin(y)$

2. $g(t) = \sqrt[4]{t^3} + \frac{7}{t^4}$

3. $f(x) = -x^5 - 4x^4 + 5x - 4$

Learning target DF4, version 2

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $h(w) = -\frac{\cos(w)}{6w^2 + 5w + 4}$

2. $g(w) = \frac{3w^2 + 5w - 1}{w^8}$

3. $f(w) = -(4w^2 + 6w + 1) \ln(w)$

Learning target DF5, version 2

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $h(x) = -9 \cos(-5x^2 + 3x + 4)$

2. $f(y) = -9 \sin(y^{7/2})$

3. $g(t) = -9 \left(\sin(t) \right)^{7/2}$

4. $k(w) = (3w + e^w - 1)^4$

Learning target DF6, version 2

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum and difference, etc.) you are using.

1. $f(w) = \left(\frac{5w^6 + 1}{5w^6 + 2} \right)^6$

2. $g(x) = (5x^5 - 2x^3)^6 \sqrt{x}$

3. $h(y) = \sqrt{\sin(-2y^4 + 4)}$